

OpenSlot

University Office-Hours Scheduling Application

CP476 INTERNET COMPUTING · WINTER 2026

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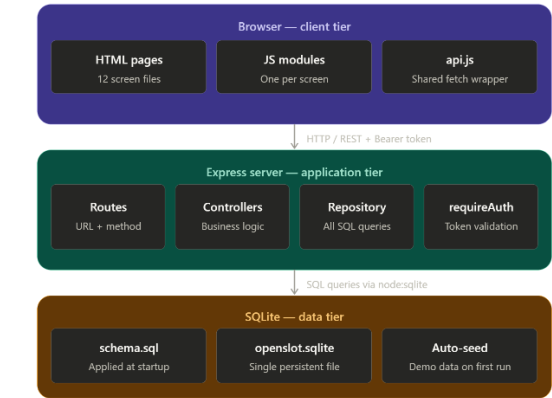
12 screens · 4 db tables · 11 api routes · 3 devs

01 ARCHITECTURE

Tier 1 — Browser (Client)
HTML5 · CSS3 · Vanilla JS ES Modules · One HTML page + JS module per screen
↓ HTTP / REST + Bearer token auth

Tier 2 — Express Server
Node.js 23 + Express 4 · Route → Controller → Repository · requireAuth middleware
↓ SQL queries via node:sqlite (DatabaseSync)

Tier 3 — SQLite Database
openslot.sqlite · WAL mode · FK constraints ON · Auto-seeds on first run



02 TECHNOLOGY STACK

Front-end	HTML5 · CSS3 · Vanilla JS ES Modules
Back-end	Node.js 23 + Express 4
Database	SQLite via node:sqlite
Auth	Bearer token (demo-token-{userId})
Pattern	Route → Controller → Repository
Dev tools	node --watch · deps: express, cors

03 STUDENT WORKFLOW

STUDENT WORKFLOW
browse-slots · confirm-booking · my-bookings · appointment-details

► **Browse & filter slots**
View all posted public slots across professors and courses

► **Book with notes**
Confirm on detail page — session notes up to 600 chars

► **Track appointments**
Dashboard summary and full bookings list with status

► **Reschedule or cancel**
Move to a new slot or cancel before session start

04 PROFESSOR WORKFLOW

PROFESSOR WORKFLOW
professor-dashboard · professor-schedule · slot-create · professor-courses

► **Create availability slots**
Date, time, mode (in-person / virtual), topic, visibility

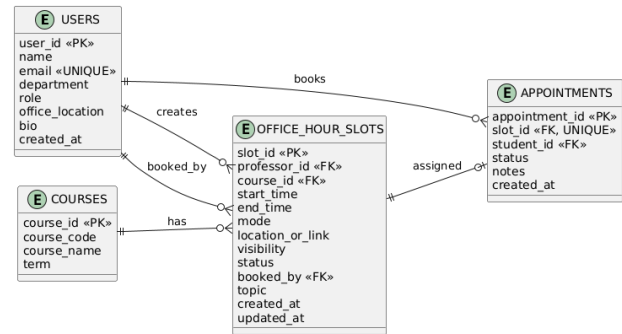
► **Dashboard overview**
Booked sessions and open slots at a glance

► **Weekly schedule view**
Slot distribution across the week per course

► **Profile & courses**
Update office location, bio, email; view course cards

05 DATABASE SCHEMA

- users — user_id · email · role · department · office_location
- courses — course_id · course_code · course_name · term
- office_hour_slots — slot_id · professor_id · start/end_time · mode · status
- appointments — appointment_id · slot_id · student_id · status · notes



06 TEAM RESPONSIBILITIES

Akif Rahman
Frontend & Project Setup Lead

- Repo, README, Kanban & data entities (#1, #2, #5)
- Tech stack decision & activity blog (#6, #23)
- Professor Dashboard UI (#19)
- Slot Creation Form (#20)
- Appointment Detail View (#21)
- Workflow Integration & UI Consistency (#22)
- CRUD ops — appointment data (#25)
- Front-end form validation (#28)
- Code review & refactoring (#36, #37)

Benjamin Okojie
Student-Facing Frontend

- Wireframes & app pages / routes outline (#4, #7)
- Student Booking Interface (#9)
- Schedule Rendering — mock data (#10)
- Slot Selection & Booking Interaction (#11)
- Navigation Between Student Screens (#12)
- UI Testing & Consistency Review (#13)
- Front-End → Backend API Integration (#24)
- Server-side input validation (#27)
- Deployment instructions & testing report (#31, #32)

Dharmik Patel
Backend & Database Lead

- User stories & acceptance criteria (#3)
- Backend skeleton & initial API stubs (#17, #18)
- SQL CREATE TABLE statements (#16)
- Relational schema design & ER diagram (#14, #15)
- Connect routes to database queries (#26)
- Appointment update & cancellation (#29)
- End-to-end workflow testing (#30)
- Architecture overview diagram (#33)

